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AI Summary

This academic essay by Salman Khan examines the greenhouse effect as both a natural, life-sustaining phenomenon and a dangerous, human-driven environmental threat. The natural greenhouse effect regulates the Earth's temperature by trapping necessary infrared radiation; however, human activities have intensified this process, leading to unprecedented global warming. Since the Industrial Revolution, the combustion of fossil fuels, widespread deforestation, intensive agricultural practices, and industrial emissions have drastically increased the concentration of greenhouse gases such as carbon dioxide, methane, and nitrous oxide in the atmosphere. The essay highlights that carbon dioxide is the primary driver of climate change, contributing approximately 66% of radiative forcing since 1750. Furthermore, it emphasizes the significant, short-term warming potential of methane, which is notably more potent than carbon dioxide. The document details how these atmospheric changes lead to severe global consequences, including rising sea levels, glacier retreat, extreme weather events, and risks to food and water security. With atmospheric CO2 reaching 420 ppm in 2023, the text underscores that climate change is an immediate reality rather than a future prediction. It concludes that mitigating this crisis requires systemic shifts, including urgent decarbonization, improvements in land management, a transition to sustainable food systems, and the preservation of natural carbon sinks like forests to stabilize the planet's thermal balance.

Keywords

Greenhouse Effect, greenhouse gases, climate, Greenhouse, Global Warming, Warming, World Bank, Department BCA, Khan KID, Bhavna Mehla